The Sherman–Morrison–Woodbury Formula and Its Applications in Data Science

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1 Introduction

Solving a linear system of the form

$$Ax = b$$

is the ultimate end point of many procedures in numerical linear algebra, scientific computing, statistics, and data science. In many applications, a complicated computational procedure eventually reduces to the problem of solving such a linear system.

If the matrix A is nonsingular, then theoretically the solution may be written as

$$x = A^{-1}b.$$

In practical computation, of course, one generally does not explicitly compute A^{-1} merely for the purpose of solving a linear system. Nevertheless, the inverse notation is extremely useful for understanding how a solution changes when the underlying problem changes.

The vector b may change from one problem to another. However, in many important applications, the matrix A itself may also change. This is particularly common in data science, where the matrices under consideration are constructed from data. New observations may be added, old observations may be removed, weights may be changed, or a model may be updated as new information becomes available.

Suppose that we have already performed the expensive computation associated with a matrix A, and then A is modified to a new matrix

$$\widetilde{A} = A + \Delta A.$$

One possibility is simply to forget everything we previously computed and solve the new problem from the beginning. However, this may be extremely wasteful, particularly when A is very large and the change ΔA possesses some simple structure.

This naturally leads to the following question:

> *If a matrix changes in a structured way, can we use what we already know about the original matrix to efficiently obtain information about the modified matrix?*

The Sherman–Morrison and Woodbury formulas provide a remarkably elegant answer to this question.

2 Rank-One Changes

The simplest structured modification of a matrix is a rank-one change.

Let

$$u, v \in \mathbb{R}^n.$$

Then the outer product

$$uv^T$$

is an $n \times n$ matrix. Provided that u and v are nonzero, this matrix has rank one.

Therefore, if the change in A can be represented as

$$\Delta A = uv^T,$$

then the modified matrix becomes

$$\widetilde{A} = A + uv^T.$$

Similarly, we may encounter

$$\widetilde{A} = A - uv^T.$$

An important point should be emphasized here. A *rank-one change* does not necessarily mean that only one entry of the matrix has changed, nor does it mean that the numerical change must be small. In fact, uv^T may change every entry of A. What matters is the structure of the change, namely,

$$\operatorname{rank}(\Delta A) = 1.$$

For example,

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}.$$

Every entry of the resulting matrix is nonzero, but the matrix still has rank one.

Thus, the number of entries that change is not the important issue. The important issue is the *rank of the change.*

3 From Rank-One Changes to General Low-Rank Changes

A change in A may, of course, have rank greater than one. Suppose

$$\operatorname{rank}(\Delta A) = k.$$

Such a change may be expressed as a sum of rank-one matrices,

$$\Delta A = u_1 v_1^T + u_2 v_2^T + \cdots + u_k v_k^T.$$

Consequently,

$$\widetilde{A} = A + u_1 v_1^T + u_2 v_2^T + \cdots + u_k v_k^T.$$

At this point there are two natural possibilities.

First, one may update the matrix one rank-one modification at a time. This leads naturally to the **Sherman–Morrison formula**.

Second, we may collect all of the rank-one modifications together. Define

$$U = \begin{bmatrix} u_1 & u_2 & \cdots & u_k \end{bmatrix}, \qquad V = \begin{bmatrix} v_1 & v_2 & \cdots & v_k \end{bmatrix}.$$

Then

$$UV^T = u_1 v_1^T + u_2 v_2^T + \cdots + u_k v_k^T,$$

and therefore

$$\boxed{\widetilde{A} = A + UV^T.}$$

This leads to the more general **Woodbury matrix identity**.

The basic philosophy behind both formulas can therefore be summarized as

$$\boxed{\text{Do not recompute everything when only a structured part has changed.}}$$

The Sherman–Morrison formula handles a rank-one modification, while the Woodbury formula extends the same idea to a low-rank modification.

In the following sections, we shall derive these formulas directly from the linear system

$$Ax = b,$$

rather than simply presenting them as identities to be memorized. This approach makes clear both where the formulas come from and why they are computationally useful.

4 The Sherman–Morrison Formula

We now consider the simplest possible low-rank modification of a matrix: a rank-one update. Suppose that $A \in \mathbb{R}^{n\times n}$ is nonsingular and that A^{-1} is already known. Let $u, v \in \mathbb{R}^n$, and suppose that A is modified according to

$$\widetilde{A} = A - uv^T.$$

Our objective is to determine the inverse of the modified matrix,

$$(A - uv^T)^{-1},$$

without computing this inverse again from the beginning.

Rather than attempting to guess the formula for the new inverse, we shall derive it directly by asking a much more natural question:

How does the solution of $Ax = b$ change when A is replaced by $A - uv^T$?

4.1 Beginning with the Modified Linear System

Consider

$$(A - uv^T)x = b.$$

Expanding the left-hand side gives

$$Ax - uv^T x = b.$$

Since A^{-1} is assumed to be known, premultiply the equation by A^{-1}:

$$A^{-1}Ax - A^{-1}uv^T x = A^{-1}b.$$

Using

$$A^{-1}A = I,$$

we obtain

$$x - A^{-1}uv^T x = A^{-1}b.$$

To simplify the notation, define

$$z = A^{-1}u$$

and

$$y = A^{-1}b.$$

Therefore,

$$\boxed{x - zv^T x = y.}$$

4.2 The Important Observation: $v^T x$ is a Scalar

At first sight, the equation

$$x - zv^T x = y$$

still contains the unknown vector x in two places. However, the second appearance of x has a very special form.

Since

$$v^T \in \mathbb{R}^{1\times n}$$

and

$$x \in \mathbb{R}^{n\times 1},$$

we have

$$v^T x \in \mathbb{R}.$$

Thus, $v^T x$ is not another unknown vector. It is merely a scalar.

Let

$$\boxed{\alpha = v^T x.}$$

Then

$$x - z\alpha = y,$$

and consequently

$$\boxed{x = y + \alpha z.}$$

This equation reveals the basic geometry of the rank-one update. The vector

$$y = A^{-1}b$$

is the solution associated with the original matrix A, while the new solution differs from y only by a scalar multiple of

$$z = A^{-1}u.$$

In other words,

$$\boxed{\text{new solution} = \text{old solution} + \text{scalar correction} \times \text{correction direction}.}$$

The entire effect of the rank-one modification has therefore been reduced to determining a single scalar α.

4.3 Solving for the Scalar α

We know that

$$x - z\alpha = y.$$

Premultiply this equation by v^T:

$$v^T x - v^T z\alpha = v^T y.$$

But by definition,

$$v^T x = \alpha.$$

Therefore,

$$\alpha - v^T z\alpha = v^T y.$$

Since $v^T z$ is also a scalar, we may factor α:

$$(1 - v^T z)\alpha = v^T y.$$

Provided that

$$1 - v^T z \neq 0,$$

we obtain

$$\boxed{\alpha = \frac{v^T y}{1 - v^T z}.}$$

Now substitute

$$z = A^{-1} u$$

and

$$y = A^{-1} b.$$

Then

$$\alpha = \frac{v^T A^{-1} b}{1 - v^T A^{-1} u}.$$

4.4 Recovering the New Solution

Recall that

$$x = y + \alpha z.$$

Substituting the expressions for y, z, and α gives

$$x = A^{-1} b + A^{-1} u \frac{v^T A^{-1} b}{1 - v^T A^{-1} u}.$$

Equivalently,

$$x = A^{-1} b + A^{-1} u \left(1 - v^T A^{-1} u\right)^{-1} v^T A^{-1} b.$$

Now factor b from the right:

$$x = \left[A^{-1} + A^{-1} u \left(1 - v^T A^{-1} u\right)^{-1} v^T A^{-1}\right] b.$$

On the other hand, the modified system was

$$(A - uv^T)x = b,$$

so

$$x = (A - uv^T)^{-1}b.$$

Since the preceding relation holds for an arbitrary vector b, we obtain

$$\boxed{(A - uv^T)^{-1} = A^{-1} + A^{-1}u\left(1 - v^T A^{-1}u\right)^{-1} v^T A^{-1}.}$$

This is the **Sherman–Morrison formula** for a rank-one subtractive update.

4.5 The More Common Additive Form

The formula is frequently written for the update

$$\widetilde{A} = A + uv^T.$$

Replacing u by $-u$ in the preceding result gives

$$\boxed{(A + uv^T)^{-1} = A^{-1} - A^{-1}u\left(1 + v^T A^{-1}u\right)^{-1} v^T A^{-1}.}$$

Since

$$1 + v^T A^{-1}u$$

is a scalar, this may also be written as

$$\boxed{(A + uv^T)^{-1} = A^{-1} - \frac{A^{-1}uv^T A^{-1}}{1 + v^T A^{-1}u}.}$$

4.6 When Does the Formula Fail?

The Sherman–Morrison formula requires

$$\boxed{1 + v^T A^{-1}u \neq 0}$$

for the additive form, or

$$\boxed{1 - v^T A^{-1}u \neq 0}$$

for the subtractive form.

This condition is not merely an algebraic inconvenience. If

$$1 + v^T A^{-1}u = 0,$$

then the updated matrix

$$A + uv^T$$

is singular, and consequently its inverse does not exist.

Thus, the denominator appearing in the Sherman–Morrison formula also provides information about whether the rank-one modification destroys the invertibility of the original matrix.

4.7 Interpretation of the Formula

The Sherman–Morrison formula may be viewed schematically as

$$\boxed{\text{new inverse} = \text{old inverse} - \text{rank-one correction.}}$$

Indeed,

$$A^{-1}u$$

is an $n \times 1$ vector and

$$v^T A^{-1}$$

is a $1 \times n$ vector. Therefore,

$$A^{-1}u\, v^T A^{-1}$$

is itself a rank-one matrix.

Hence, a rank-one modification of A produces a rank-one correction to A^{-1}, provided the modified matrix remains nonsingular.

This is the essential computational idea behind the formula. If the information associated with A has already been computed, then a structured change in A should, whenever possible, be handled by an efficient update rather than by discarding the previous computation and starting again.

5 A Numerical Example of the Sherman–Morrison Formula

We now illustrate the Sherman–Morrison formula using a simple 2×2 example.

Let

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}, \qquad u = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

We consider the subtractive rank-one update

$$\widetilde{A} = A - uv^T.$$

First compute

$$uv^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}.$$

Therefore,

$$\widetilde{A} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix}.$$

Our objective is to compute

$$\widetilde{A}^{-1} = (A - uv^T)^{-1}$$

using the Sherman–Morrison formula.

For the subtractive case,

$$(A - uv^T)^{-1} = A^{-1} + A^{-1}u \left(1 - v^T A^{-1} u\right)^{-1} v^T A^{-1}.$$

5.1 Step 1: Compute A^{-1}

Since A is diagonal,

$$A^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix}.$$

5.2 Step 2: Compute $A^{-1}u$

We obtain

$$A^{-1}u = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix}.$$

5.3 Step 3: Compute $v^T A^{-1}$

Similarly,

$$v^T A^{-1} = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \end{bmatrix}.$$

5.4 Step 4: Compute the Scalar Denominator

We next compute

$$v^T A^{-1} u.$$

Using the result from the previous step,

$$v^T A^{-1} u = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} = \frac{1}{2}.$$

Therefore,

$$1 - v^T A^{-1} u = 1 - \frac{1}{2} = \frac{1}{2},$$

and hence

$$\left(1 - v^T A^{-1} u\right)^{-1} = 2.$$

5.5 Step 5: Compute the Rank-One Correction

The correction term is

$$A^{-1}u \left(1 - v^T A^{-1} u\right)^{-1} v^T A^{-1}.$$

Substituting the quantities already computed gives

$$\begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} (2) \begin{bmatrix} 1/2 & 0 \end{bmatrix}.$$

Since

$$2 \begin{bmatrix} 1/2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

the correction becomes

$$\begin{bmatrix} 1/2 \\ 1/3 \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 \\ 1/3 & 0 \end{bmatrix}.$$

5.6 Step 6: Obtain the Updated Inverse

Therefore,

$$(A - uv^T)^{-1} = \begin{bmatrix} 1/2 & 0 \\ 0 & 1/3 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 1/3 & 0 \end{bmatrix}.$$

Thus,

$$\boxed{(A - uv^T)^{-1} = \begin{bmatrix} 1 & 0 \\ 1/3 & 1/3 \end{bmatrix}.}$$

5.7 Direct Verification

We can verify the result directly by multiplication:

$$\begin{bmatrix} 1 & 0 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1/3 & 1/3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Hence,

$$\boxed{\widetilde{A}\widetilde{A}^{-1} = I,}$$

confirming that the Sherman–Morrison formula gives the correct inverse.

5.8 What This Example Shows

The important point is not merely that the arithmetic works. The example illustrates the structure of the update:

$$\boxed{\text{new inverse} = \text{old inverse} + \text{rank-one correction.}}$$

We did not need to derive the inverse of the modified matrix independently. Instead, we used the inverse already known for A and corrected it using only the vectors u and v.

For a 2×2 matrix this may not appear impressive. However, when A is extremely large and the modification has low rank, this same idea can save a substantial amount of computation.

6 The Woodbury Matrix Identity

The Sherman–Morrison formula handles a rank-one update of the form

$$A + uv^T.$$

Suppose now that the change in A has rank greater than one. Let

$$U, V \in \mathbb{R}^{n \times k},$$

where typically

$$k \ll n.$$

Then a rank-k modification may be written in the form

$$\widetilde{A} = A + UV^T.$$

The objective is again to determine the inverse of the modified matrix without recomputing everything from the beginning.

Rather than memorizing the Woodbury formula, we shall derive it directly from the modified linear system

$$(A + UV^T)x = b.$$

6.1 Beginning with the Modified System

Consider

$$(A + UV^T)x = b.$$

Expanding gives

$$Ax + UV^T x = b.$$

Assuming that A^{-1} is already known, premultiply by A^{-1}:

$$A^{-1}Ax + A^{-1}UV^T x = A^{-1}b.$$

Therefore,

$$x + A^{-1}UV^T x = A^{-1}b.$$

Define

$$Z = A^{-1}U$$

and

$$y = A^{-1}b.$$

Then

$$\boxed{x + ZV^T x = y.}$$

6.2 The Key Difference from Sherman–Morrison

In the Sherman–Morrison derivation, the quantity

$$v^T x$$

was a scalar.

Here,

$$V^T x$$

is generally not a scalar.

Since

$$V^T \in \mathbb{R}^{k\times n}$$

and

$$x \in \mathbb{R}^{n\times 1},$$

we have

$$V^T x \in \mathbb{R}^{k\times 1}.$$

Thus, instead of introducing one scalar unknown, we introduce a small vector.

Let

$$\boxed{\alpha = V^T x,}$$

where

$$\alpha \in \mathbb{R}^k.$$

Then

$$x + Z\alpha = y,$$

and consequently

$$\boxed{x = y - Z\alpha.}$$

The original unknown vector x may have dimension n, but the correction has now been reduced to determining only the k components of α.

6.3 Solving for the Correction Vector

Starting from

$$x + Z\alpha = y,$$

premultiply by V^T:

$$V^T x + V^T Z\alpha = V^T y.$$

Since

$$V^T x = \alpha,$$

we obtain

$$\alpha + V^T Z\alpha = V^T y.$$

Factor α:

$$(I + V^T Z)\alpha = V^T y.$$

Provided that

$$I + V^T Z$$

is nonsingular, we obtain

$$\boxed{\alpha = (I + V^T Z)^{-1} V^T y.}$$

Now substitute

$$Z = A^{-1} U$$

and

$$y = A^{-1} b.$$

Then

$$\boxed{\alpha = \left(I + V^T A^{-1} U\right)^{-1} V^T A^{-1} b.}$$

6.4 Recovering the New Solution

Recall that

$$x = y - Z\alpha.$$

Substituting the expressions above gives

$$x = A^{-1} b - A^{-1} U \left(I + V^T A^{-1} U\right)^{-1} V^T A^{-1} b.$$

Factor b from the right:

$$x = \left[A^{-1} - A^{-1} U \left(I + V^T A^{-1} U\right)^{-1} V^T A^{-1}\right] b.$$

On the other hand,

$$x = (A + UV^T)^{-1} b.$$

Therefore,

$$\boxed{(A + UV^T)^{-1} = A^{-1} - A^{-1} U \left(I + V^T A^{-1} U\right)^{-1} V^T A^{-1}.}$$

This is the **Woodbury matrix identity**.

6.5 Why the Formula is Computationally Attractive

Suppose

$$A \in \mathbb{R}^{n\times n},$$

while

$$U, V \in \mathbb{R}^{n\times k}.$$

Then

$$V^T A^{-1} U$$

has dimension

$$(k \times n)(n \times n)(n \times k) = k \times k.$$

Therefore,

$$I + V^T A^{-1} U$$

is only a $k \times k$ matrix.

This is the central computational advantage of the Woodbury formula.

If

$$n = 100000$$

but

$$k = 10,$$

then the matrix appearing in the middle inverse has size only

$$10 \times 10.$$

Thus, a modification to a very large matrix can sometimes be handled by solving a much smaller problem.

6.6 Relationship with Sherman–Morrison

The Sherman–Morrison formula is contained inside the Woodbury identity as the special case

$$k = 1.$$

If

$$U = u$$

and

$$V = v,$$

then

$$V^T A^{-1} U = v^T A^{-1} u$$

is a scalar.

Hence,

$$I + V^T A^{-1} U$$

reduces to

$$1 + v^T A^{-1} u.$$

Therefore the Woodbury formula becomes

$$(A + uv^T)^{-1} = A^{-1} - A^{-1}u\left(1 + v^T A^{-1} u\right)^{-1} v^T A^{-1},$$

which is precisely the Sherman–Morrison formula.

Thus,

$$\boxed{\text{Sherman–Morrison = rank-one Woodbury.}}$$

Equivalently,

$$\boxed{\text{Woodbury = Sherman–Morrison generalized to rank } k.}$$

6.7 The Main Idea

The Woodbury identity may be viewed schematically as

$$\boxed{\text{new inverse = old inverse − low-rank correction.}}$$

The essential lesson is therefore the same as before:

$$\boxed{\text{Exploit the structure of the change instead of recomputing everything.}}$$

When the modification has low rank relative to the size of the original matrix, this principle can lead to a substantial reduction in computational work.

7 Conclusion

The Sherman–Morrison and Woodbury formulas are more than convenient identities for updating the inverse of a matrix. They illustrate a much broader principle in numerical computation:

$$\boxed{\text{When a problem changes, we should first ask how much of the old computation can still be used.}}$$

In numerical linear algebra and data science, we frequently encounter systems of the form

$$Ax = b.$$

Neither the data nor the mathematical objects constructed from the data remain unchanged forever. New observations may arrive, old observations may be removed, parameters may change, or the underlying model itself may require an update. Consequently, both b and A may change as a computation evolves.

The immediate reaction to such a change should not necessarily be to discard the previous computation and start again.

Instead, before starting the machine, one should first stop and examine the mathematical structure of the new problem.

Suppose

$$A \longrightarrow A + \Delta A.$$

Before recomputing everything, we should ask:

$$\text{What is the structure of } \Delta A?$$

Can it be represented as a rank-one update,

$$\Delta A = uv^T?$$

Can it be represented as a sum of a small number of rank-one updates,

$$\Delta A = u_1 v_1^T + \cdots + u_k v_k^T?$$

Or, equivalently, can it be represented as a low-rank update,

$$\Delta A = UV^T, \qquad k \ll n?$$

If the answer is yes, then the Sherman–Morrison or Woodbury formula may allow us to reuse information from the original problem rather than performing the entire computation again.

Thus, the important question is not simply

How do I compute the solution of the new problem?

but rather

How is the new problem related to the problem that I have already solved?

This change in viewpoint is extremely important. A computer can perform an enormous number of arithmetic operations, but the availability of computational power should not replace mathematical reasoning. There is no need to immediately begin a large computation merely because a system has changed. We should first determine whether the change possesses a structure that allows the task to be reduced.

The Sherman–Morrison formula teaches us how to exploit a rank-one change:

$$\boxed{\text{rank-one change} \quad \Longrightarrow \quad \text{rank-one correction.}}$$

The Woodbury identity extends the same philosophy:

$$\boxed{\text{low-rank change} \quad \Longrightarrow \quad \text{low-dimensional correction problem.}}$$

For large-scale data science problems, this distinction can be substantial. If a matrix has dimension n but its modification has rank k, where

$$k \ll n,$$

then the Woodbury identity can transfer the essential new computation to a much smaller $k \times k$ system.

The broader lesson therefore extends far beyond these two formulas:

Before recomputing a problem from the beginning, examine what actually changed. Preserve what can be preserved, exploit the structure of the change, and reduce the new computation whenever possible.

In numerical mathematics, the most efficient computation is often not the computation performed fastest by the machine, but the computation that mathematical reasoning allows us to avoid altogether.